A. Appendix

Table 6. Theoretical Power Consumption Comparison. The power consumption is estimated by using the model proposed by Lv et al. (2024).

Environment	Model	Number of Floating Point Operations	Power	Energy Reduction
Atari Enviroment (CNN-based Architecture)	DQN	$9,303,552$	$116\mu J$	$-$
	DSQN	$432,260$	$2.7\mu J$	$43\times\downarrow$
	DTSQN	$432,260$	$2.7\mu J$	$43\times\downarrow$
	DATSQN	$432,260$	$2.7\mu J$	$43\times\downarrow$

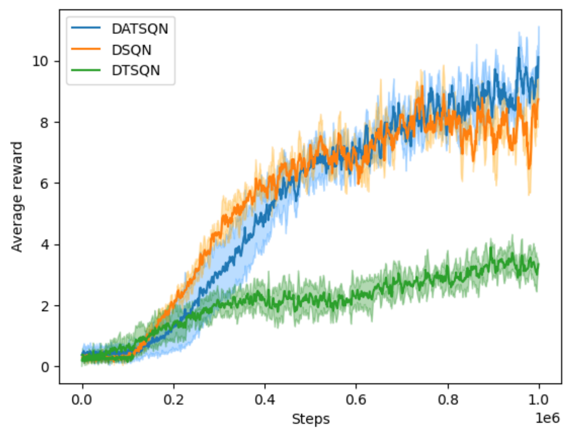

Figure 5. Average reward achieved by each RL agent across 3 different seeds (0, 10, 42) in the Breakout environment.

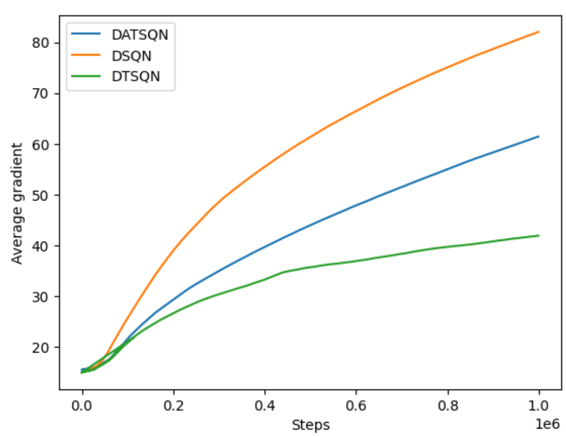

Figure 6. The average norm of the gradients across the network during training for each RL agent, evaluated over 3 different seeds in the Breakout environment.

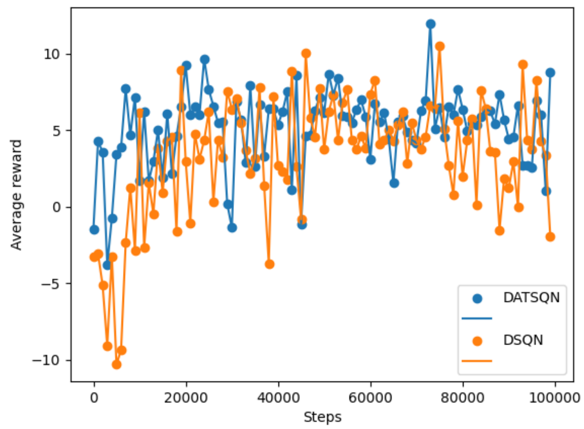

Figure 7. Average reward comparison of DSQN and DATSQN in the Highway environment using the Double DQN architecture proposed by Lu et al.

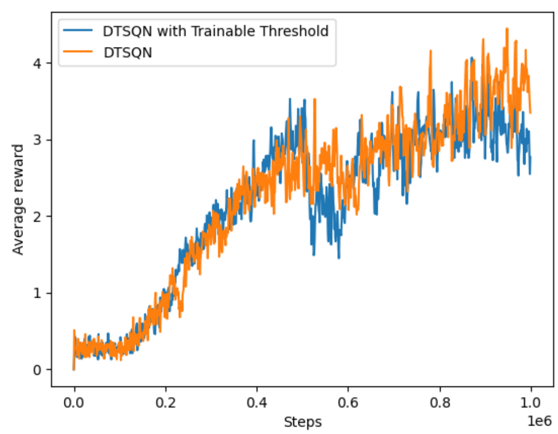

Figure 8. Average reward achieved by DTSQN with fixed and trainable thresholds in the Breakout environment.